

SUMMARY

Data-driven Supply Chain Analyst with 3 years of experience optimizing demand forecasting, supply planning, and inventory management across healthcare and global technology organizations. Proven expertise in advanced analytics, machine learning, and simulation modeling to drive cost reduction, service-level improvements, and risk-informed decision-making. Strong background in Python, SQL, Power BI, and statistical forecasting, with hands-on experience supporting S&OP / IBP cycles in fast-paced, regulated environments. Recognized for analytical rigor, executive-level communication, and continuous improvement aligned with Lean Six Sigma best practices.

SKILLS

- **Programming & Analytics:** Python (Pandas, NumPy, Scikit-learn, Statsmodels), R (Forecasting, Statistical Modeling), SQL (Advanced Queries, Optimization), Excel (Advanced Formulas, VBA, Macros)
- **Visualization & BI:** Power BI (DAX, Data Modeling, Executive Dashboards)
- **Forecasting & ML:** Time Series Forecasting (ARIMA, SARIMA), Machine Learning (Random Forest, XGBoost), Forecast Accuracy (MAPE, RMSE, Bias)
- **Supply Chain & IE Tools:** Demand Forecasting, Supply Planning, Inventory Optimization (EOQ, Safety Stock, Reorder Points), Supplier Performance & Risk Analysis, S&OP / IBP, Lean Six Sigma, Kaizen, FMEA, APQP
- **Cost & Decision Analysis:** Cost-Benefit Analysis, NPV, Life-Cycle Costing (LCC), Earned Value Management (EVM), Decision Trees, Monte Carlo Simulation, Scenario Planning
- **Systems & Simulation:** Arena (Discrete-Event Simulation), SysML, MBSE, AutoCAD, SolidWorks, Ansys

EXPERIENCE

UnitedHealth Group Supply Chain Analyst

June 2025 – Present

- Lead demand forecasting and inventory analytics for mission-critical healthcare supply categories across multiple distribution networks.
- Developed Python-based forecasting models (ARIMA, Random Forest, XGBoost), improving forecast accuracy by 18–22% and reducing stockouts and excess inventory.
- Designed executive-level Power BI dashboards integrating demand trends, inventory turns, supplier KPIs, and service levels to support real-time decision-making.
- Optimized safety stock and reorder point policies using statistical models, reducing carrying costs while maintaining 99% service levels.
- Conduct scenario planning and Monte Carlo simulations to assess demand volatility, supply risk, and supplier disruptions in regulated environments.
- Support S&OP / IBP cycles by aligning demand signals with capacity, sourcing, and supplier constraints.
- Performed cost-benefit and NPV analyses for sourcing and inventory strategies, delivering measurable cost savings.
- Automated SQL and Python data pipelines, improving reporting efficiency and reducing manual effort by 30%+.
- Applied Lean Six Sigma and Kaizen methodologies to identify process inefficiencies and drive continuous improvement initiatives.

Dell Technologies

October 2021 – December 2023

Supply Chain Analyst (Supply Planning & Forecasting)

- Supported global supply planning and demand forecasting for high-volume technology products across APAC and North America.
- Built Excel VBA and SQL-based planning models to improve demand-supply alignment and inventory health.
- Implemented statistical forecasting models (ARIMA, trend and seasonality analysis) to improve demand visibility and reduce forecast bias.
- Collaborated cross-functionally with manufacturing, procurement, and logistics teams to execute capacity planning and inventory optimization strategies.
- Developed Power BI dashboards to track forecast accuracy, inventory turns, supplier performance, and fulfillment KPIs.
- Conducted EOQ and safety stock analysis, reducing excess inventory while protecting customer service commitments.
- Performed supply risk analysis and FMEA to identify potential disruptions and define mitigation strategies.
- Supported S&OP planning cycles with scenario-based insights to balance demand volatility and production constraints.
- Delivered actionable insights to leadership through data storytelling and executive-ready presentations.

EDUCATION/ CERTIFICATIONS

- Master's of Science in Industrial Systems Engineering, University of Oklahoma, Norman, OK
- Six Sigma Green Belt – Institute of Industrial & Systems Engineers (IISE) | Certificate ID: 159080946
- Advanced MBSE & SysML (2025) | Certificate ID: CERT-44365608-2